

Introduction to spatial transcriptomics

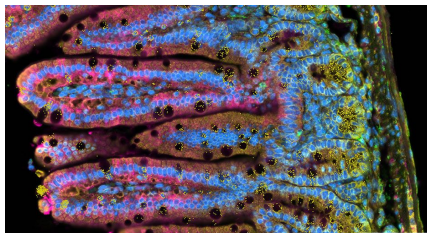
The Data Lab

What is spatial transcriptomics?

- Spatial transcriptomics measures both gene expression and its location within a tissue
- This preserves the biological context lost in single-cell methods where tissue is dissociated
- Spatial transcriptomics allows us to interrogate...
 - Where specific genes are expressed across the tissue
 - Where specific cell types are located
 - Tissue neighborhoods and niches
 - Cell-cell interactions, communication, and signaling
 - Compositional relationships among cell types including e.g., within the tumor microenvironment

Two families of spatial transcriptomics technologies

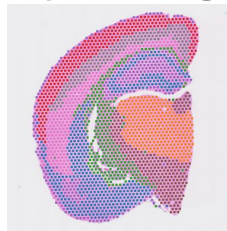
Image-based ST



10x Xenium, MERFISH, CosMX

- High, ~subcellular resolution with a targeted panel of genes, not the whole transcriptome
- Fluorescent probes directly hybridize to gene transcripts in the tissue
- Multiple rounds of imaging perform the sequencing *in situ*

Sequencing-based ST

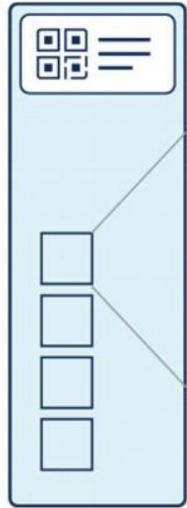


10x Visium & Visium HD, Slide-seq

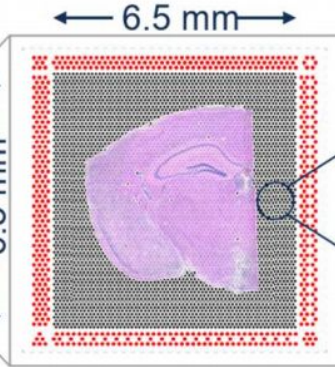
- (Historically) Spot-based lower resolution, but across a ~whole transcriptome
 - Newer methods have introduced high-resolution bins
- For older methods, RNA is captured by barcoded oligos that grab the polyA tail
- Current Visium & Visium HD are probe-based, but still essentially capture the whole transcriptome
- cDNA libraries are generated and sequenced

10x Visium slide and capture area

Visium Spatial
Gene Expression
Slide

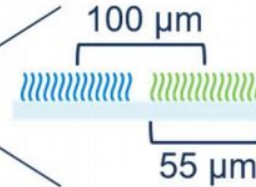


Capture Area with
~5000 Barcoded
Spots

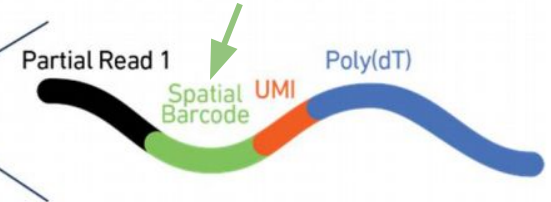


The **fiducial frame** is how the tissue gets correctly aligned to the barcode grid

Visium Gene
Expression Barcoded
Spots

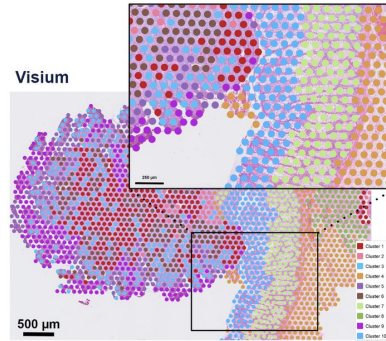


The barcode will map your reads to the location on the slide



For CytAssist, larger slides with ~14,000 spots and an 11x11 capture area are also available

**Visium v1,
fresh-frozen tissue**
Sequenced with polyA protocol



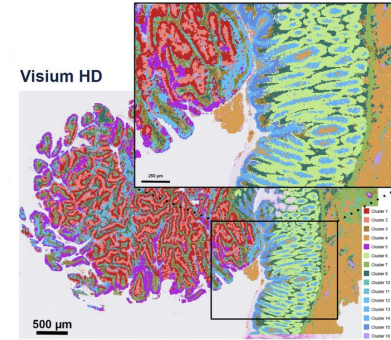
2021

**Visium v1,
FFPE**

Introduced **probe-based capture** for fixed tissue. Probe-based protocols become the standard at this point.

Visium CytAssist
CytAssist instrument decouples histology from the barcoded Visium slide

2022



2024

Visium HD

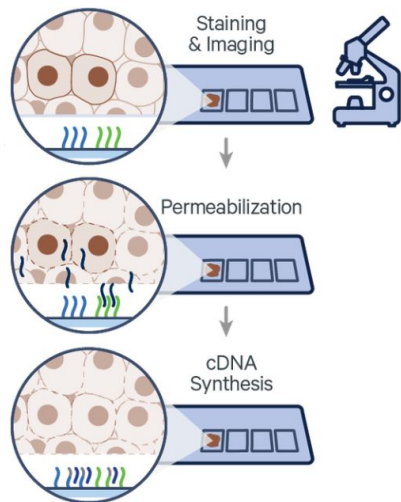
Uses CytAssist & achieves 2 µm capture resolution, vs Visium 55 µm

Note: Visium HD 3' also exists

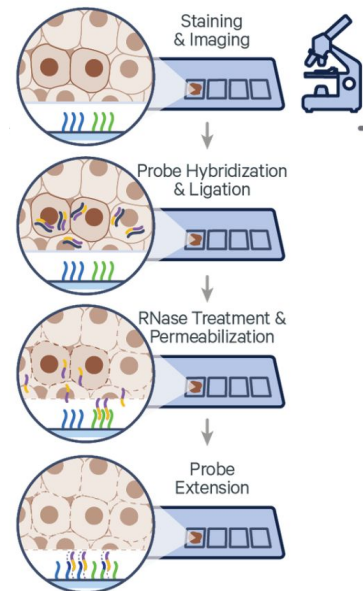
<https://www.10xgenomics.com/blog/your-introduction-to-visium-hd-spatial-biology-in-high-definition>

Library preparation for Visium v1 all happens on the Visium slide

Barcoding & cDNA Synthesis



Hybridization, Ligation & Barcoding



Fresh frozen samples:

- polyA protocol
- Whole transcriptome, but 3' bias

FFPE samples:

- Probe-based protocol
- Smaller set of genes, but no 3' bias

2022: Enter the CytAssist instrument

Steps performed on the glass slide

CytAssist machine (not you!)
transfers ligated probes to the
Visium slide

	Sample Preparation	Staining & Imaging		Probe Hybridization & Ligation	Probe Release, Extension & Elution		Library Construction	
FFPE	FFPE Tissue Sections (human or mouse) on glass slide  STOP	Deparaffinization  ~3 h	Immunofluorescence (IF) or Hematoxylin & Eosin (H&E)  ~3 h STOP	Decrosslinking  ~1.5 h	Whole Transcriptome Probe-Mediated mRNA Detection  Overnight STOP	Probe Release & Capture  ~1 h	Probe Extension, Elution & Transfer to Tube  ~1 h STOP	Pre-amplification, qPCR, SI-PCR & QC  ~4 h STOP
Fixed Frozen	FxF Tissue Sections (mouse) on glass slide  STOP	Rehydration  ~0.5 h	Hematoxylin & Eosin (H&E)  ~1 h STOP	Decrosslinking  ~1 h				
Fresh Frozen	FF Tissue Sections (human or mouse) on glass slide  STOP	Methanol Fixation  ~0.5 h	Hematoxylin & Eosin (H&E)  ~1 h STOP					

Visium gives you a gene by **spot** count matrix

- Spots can have a range of cells depending on the tissue type, so each is likely a *heterogeneous combination* of cell types
 - You can expect 1–20 cells per spot, depending on the type of tissue
 - Visium HD is genuinely one cell per spot ("bin") and may even be subcellular
- Compared to single-cell, Visium matrices are much less sparse
 - Pooling multiple cells results in higher UMI counts and more detected genes
 - Note that Visium HD matrices are still pretty sparse!



Comparing sources of technical variation

Single-cell

- Capture efficiency
- Contamination from ambient RNA
- Empty droplets
- Doublets
- Sparsity leads to ambiguous 0s (not expressed, or not sequenced?)
- Stress from dissociation

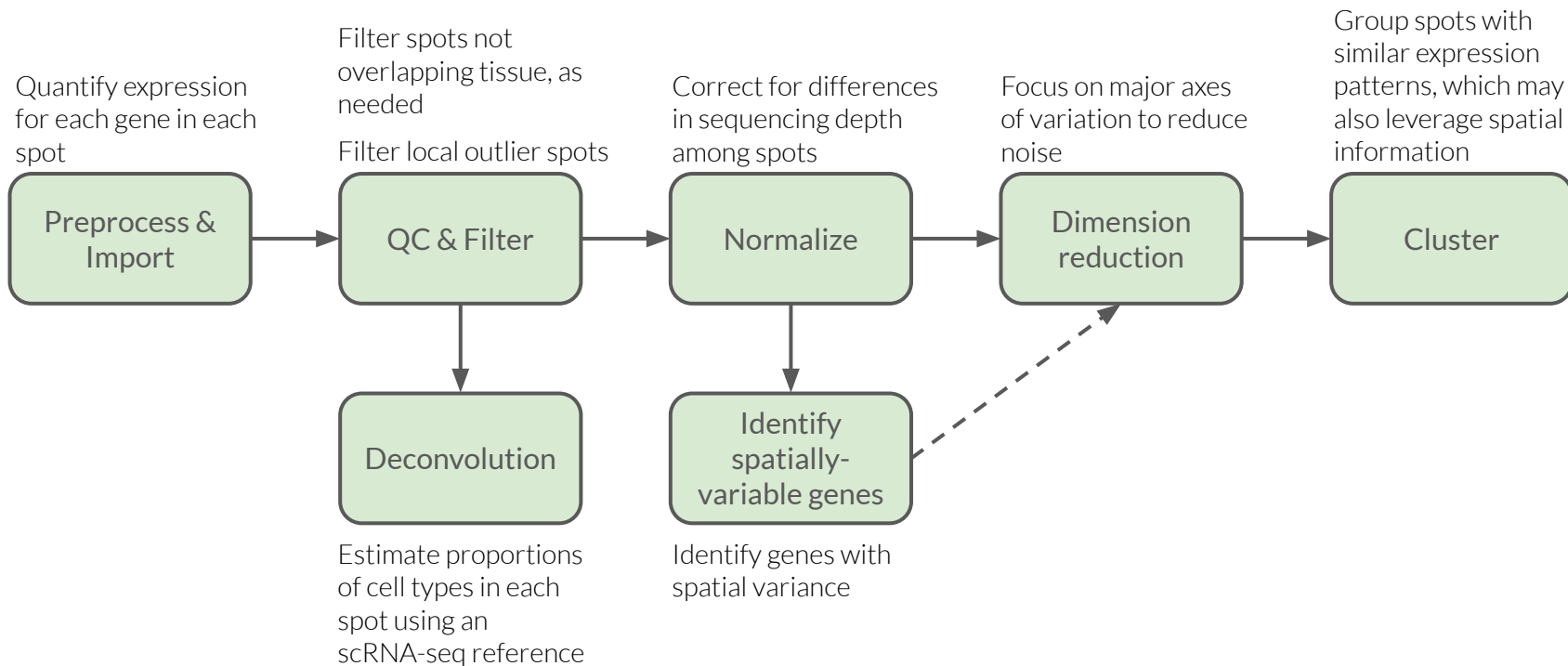
Visium

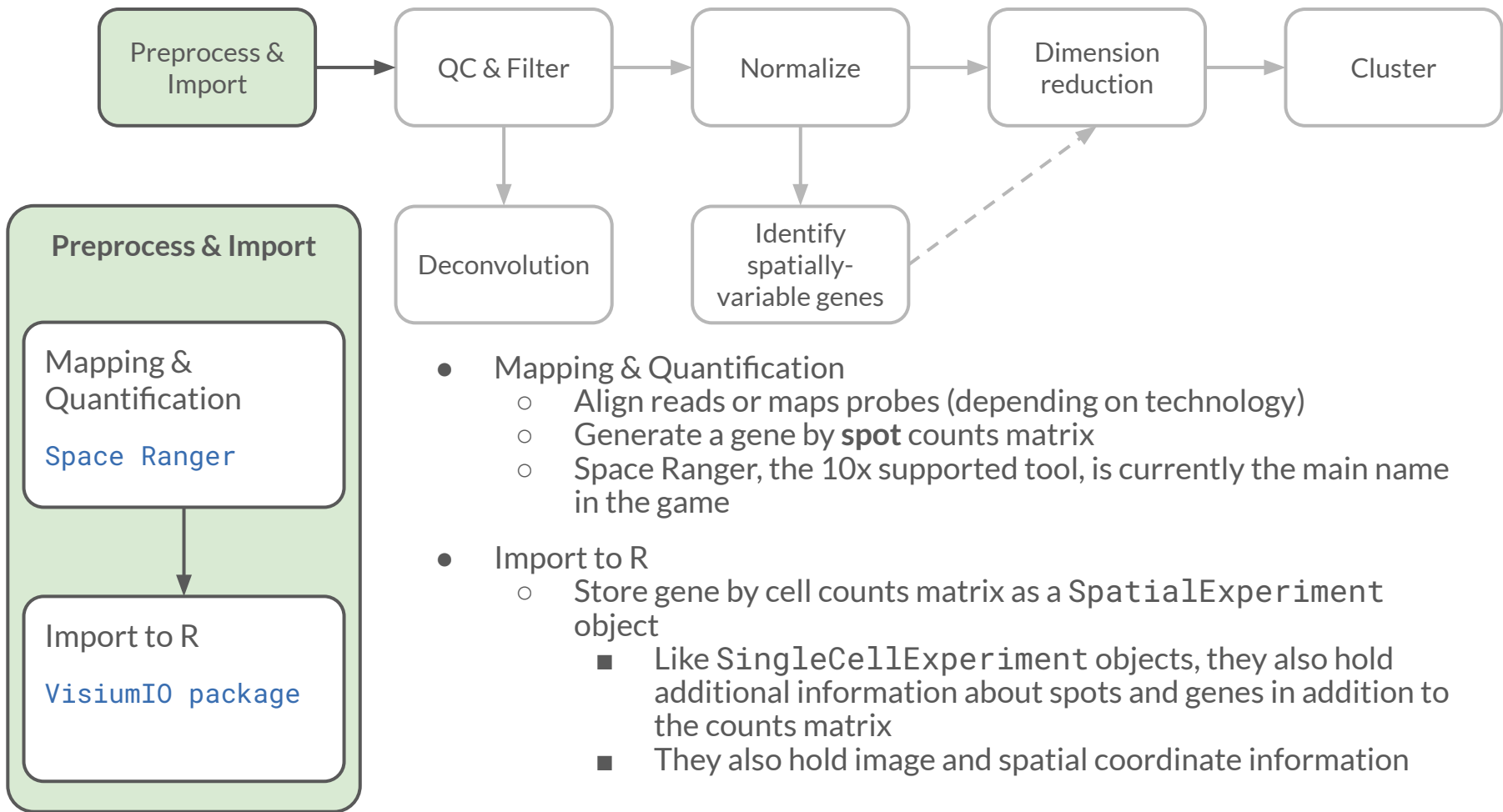
- Capture efficiency
- RNA diffusion following permeabilization
- Probe panel bias
- Tissue section artifacts (tears, folds)
- Image registration (aka alignment) error

We'll use different QC/filtering approaches for Visium to match the distinct kinds of technical variation we expect



Visium spatial transcriptomics overview





The Bioconductor **SpatialExperiment** object

